

**NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**SUBJECT: Foundations of Computer Security**  
**Mid-Semester Examination for B. Tech. Only**

■ CS 6301 ■ Duration 120 minutes ■ Full Marks 30 ■ Date: 26-September-2024■

*Answer ALL Questions*

*All parts of the question must be answered at one place.*

*Number within square bracket indicate full mark for that question.*

1. (a) Determine the number of keys in an Affine Cipher over  $\mathbb{Z}_m$  for  $m = 30, 100$ , and  $1225$ . [3]  
(b) Suppose that  $\pi$  is the following permutation of  $1, \dots, 8$ :  
$$x = \begin{array}{c|c|c|c|c|c|c|c} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \hline \pi(x) = & 4 & 1 & 6 & 2 & 7 & 3 & 8 & 5 \end{array}$$
  
Compute the permutation  $\pi^{-1}$ . Decrypt the following ciphertext, for a Permutation Cipher with  $m = 8$ , which was encrypted using the key [3]  
TGEEMNELNNTDROEOAAHDOETCSHAEIRLM.  
(c) The ciphertext *ELNI* was encrypted by Hill cipher with a  $2 \times 2$  matrix. The plain text is *dont*. Find the key. Suppose the ciphertext is *ELNK* and the plaintext is still *dont*. [2]  
Find the encryption matrix.  
(d) Find all solutions to the set of linear equations [2]  
$$3x + 2y \equiv 5 \pmod{7}$$
$$4x + 6y \equiv 4 \pmod{7}$$
2. (a) The ciphertext UCR was encrypted using the affine function  $9x + 2 \pmod{26}$ . [2]  
Find the plaintext.  
(b) Find all solutions of  $12x \equiv 28 \pmod{236}$ . [2]  
(c) With a key of "HELLO", a message of "WELCOME", with an additional key of "TEST", find the cipher text using Nihilist cipher. [2]  
(d) With neat diagram explain the key generation process for AES-192. [4]
3. (a) Discuss various cryptanalysis attacks with suitable diagram. [2]  
(b) Use the extended Euclidean algorithm to find the multiplicative inverse of the polynomial  $x^5 + x^4 + x^3 + x$  in  $\text{GF}(2^8)$  using the modulus  $x^8 + x^4 + x^3 + x + 1$ . [2]  
(c) Name three different families of algebraic groups that are commonly used in cryptographic applications and explain their properties. [2]  
(d) With suitable block diagram explain the working principle of DES. What do mean by Confusion and Diffusion? Discuss the weaknesses of DES. [4]

\*\*\*\*\*Wish you all the best\*\*\*\*\*